

Assessing Sleep Quality in Commercial Maritime Crewmembers:

Is the Sleep Disturbance Scale a Valid Measure?

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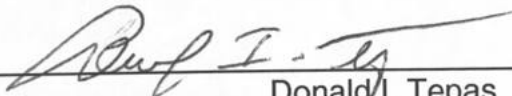
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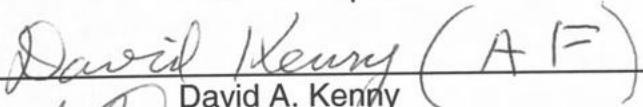
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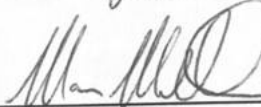
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Introduction

In the maritime industry, sleep problems can have a major impact on performance. One way to measure these sleep problems is through self-report measures. Howarth, Pratt and Tepas (1999) examined self-report survey items of sleep problems in two different maritime crews in an effort to measure sleep quality. They found on-duty sleep length to be significantly shorter than off-duty sleep length for two different types of duty in commercial maritime crews. Differences between on-duty and off-duty survey items regarding sleep problems were not significant, however. Sleep problem items that loaded on the same factor were related to an individual's quality of sleep, and so was called the Sleep Disturbance Scale. The correlation between the Sleep Disturbance Scale and sleep length was not statistically significant, however. The means of the responses for each item in the Sleep Disturbance Scale were below the midpoint of the scale for that item. The fact that the items in the Sleep Disturbance Scale were related to each other in two different samples, however, does not mean that it is a valid measure of sleep quality. Even though Howarth et al.'s (1999) findings may have been reliable, the validity of the scale comes into question because the full range of the scale was not used in either sample in that study. The purpose of the current study is to assess the validity of the Sleep Disturbance Scale by assessing its relationship with other measures related to sleep behavior.

In addition to work schedules, personality dimensions and individual differences, like "neuroticism", "extraversion" and "sleep schedule flexibility", may be related to sleep behavior. Rutenfranz (1982) described a link between individual differences in adaptation to shiftwork with differences in personality

structure. Åkerstedt and Torsvall (1981), Bruusgaard (1969), and Thiis-Evensen (1969) identified "neuroticism", an individual difference measure of mood, as an important factor in attitudes towards shiftwork. They did not show, however, whether neuroticism is a cause or a consequence of shiftwork problems. Similarly, Colquhoun and Folkard (1978) showed that "neurotic extraverts" display the greatest degree of body temperature adaptation to shiftwork and are therefore better choices for people to do shiftwork. Patkai (1971) found that there is a statistically significant relationship between "morning-ness" and "introversion" and "evening-ness" and "extraversion". Rutenfranz (1982) suggested that possibly shiftwork should be delegated based on personality type. More specifically, he discusses since "extraverts" seem to adapt their body temperatures to shift changes well and are therefore more alert, "extraverts" might be a better choice for shiftwork. Åkerstedt and Torsvall (1981) called for research to clarify the mechanism behind individual differences in adjusting to shiftwork.

Folkard, Monk, and Lobban (1979) studied a measure of an individual's ability to adjust to changing sleep schedules. They found that nurses scoring high on questions regarding "flexibility of sleeping habits" reported an inability to sleep at unusual times, and a preference for maintaining regular sleep times. They concluded that nurses scoring high on this factor might be thought of as "rigid sleeping types" and nurses scoring low would be "flexible sleeping types". They also found scores on this factor to be highly correlated with other individual difference measures of personality. Folkard et al. (1979) argued that an individual's "flexibility of sleeping habits" might be an important factor in determining adjustment to shiftwork. Costa, Lievore, Casaletti, Gaffuri, and Folkard (1989) conclude that "flexibility of sleeping habits" is one of the personality factors that is associated with better long-term tolerance to shiftwork.

Chronic fatigue is considered a long-term effect of shiftwork (Barton et al., 1995). Chronic fatigue is a, "general tiredness and lack of energy irrespective of whether an individual has not had enough sleep or has been working hard, which persists even on rest days and holidays" (Verhaegen, Maasen, & Meers, 1981). This definition of chronic fatigue should not be confused with Chronic Fatigue Syndrome, which is a medical diagnosis that does not have universal acceptance by the medical profession and requires an extensive medical diagnosis. Barton et al. (1995) devised a measure of chronic fatigue as part of the Standard Shiftwork Index (Barton et al., 1995), which is a battery of questionnaires for assessing problems relating to sleep behavior and shiftwork. In the development of the Standard Shiftwork Index, Barton et al. (1995) proposed a theoretical model showing the temporal effects of shift systems on individual differences, disturbed sleep, acute mood effects, chronic mental effects, and their resultant physical health and safety problems (See Figure 1 in Appendix A).

These cross-sectional studies are examples showing a relationship between inter-individual differences and sleep behavior. There are also examples of studies that *reject* many inter-individual difference effects on sleep. In a longitudinal study of 604 shiftworkers, Kaliterna et al. (1995) showed that after a few years of working on shifts, individual difference measures did not predict subsequent health problems in shiftworkers. In a 6-year, longitudinal study of sailors in the United States Navy, Johnson and Spinweber (1982) found no differences between good and poor sleepers in demotions, absences, or discharges. Tepas, Pratt and Howarth (2000) argue that in studying the relationship between sleep behaviors and individual differences, a clear distinction needs to be made between inter- and intra-individual difference variables.

Current Study

Maritime workshifts, or watches, are a special form of shiftwork. Typically, watchkeeping is done in 4-on, 8-off watches wherein crewmembers work for four hours, have eight hours off, work another four hours, and then have another eight hours off (Knauth et al., 1986). This type of watch splits crewmembers' sleep schedules so that crewmembers usually have two or more sleep periods in a day. This sleep behavior might lead to sleepiness while on-duty.

The Howarth et al. (1999) finding of no relationship between sleep quality, as measured by the Sleep Disturbance Scale, and sleep length in a sample of maritime crews can be interpreted in many ways. One possible interpretation is that, for the samples studied, maritime duty had an insignificant impact on sleep quality as measured by the Sleep Disturbance Scale. Another possible interpretation is that the Sleep Disturbance Scale is not a significantly valid measure of sleep quality, and therefore no maritime duty impact was found for the samples in that study.

The aim of the present research is to assess if the Sleep Disturbance Scale is a valid measure of sleep quality in one of the samples of commercial maritime crewmembers from the Howarth et al. (1999) study. To accomplish this, the relationship between the Sleep Disturbance Scale (a subjective measure of sleep quality) and several individual difference measures of sleepiness and sleep behavior in the commercial maritime crewmembers are analyzed. This relationship will be assessed by correlating sleep quality with individual difference measures of neuroticism and extraversion, flexibility/rigidity of sleeping habits, chronic fatigue, and responses to individual questions regarding sleepiness, fatigue, physical fitness, seasickness, and actual sleep length. It should be clearly noted that *no* suggestion is being made that maritime duty impacts sleep quality, that the Sleep Disturbance Scale is the only way to measure sleep

quality, that sleep quantity differences require concomitant sleep quality differences, or that the sample studied here is representative of the entire maritime population.

Data were collected from a sample of convenience of one hundred thirty-five crewmembers in the maritime industry. The crewmembers came from commercial maritime crews. The crews operated cargo ships that ranged in length. Most ships were approximately 180,000 deadweight tons. The ships performed oil transport. Two ships were company ships that carried hazardous materials (i.e., cars, food, hazardous materials, etc.). Most of the ships would arrive at a port every five to nine days. The ships were affected by weather, schedule changes, and mechanical difficulties. A typical crew consisted of 24 to 28 crewmembers per vessel. Participation was voluntary. For most cases, 75% of the population was sampled. Data were collected between March 1995 and April 1996 from different people on board. None of the ships crossed the International Date Line.

The maritime crewmembers' mean age was 41.4 years ($SD = 12.1$), their mean time in their profession was 15.9 years ($SD = 10.5$), and their mean time at their current watch was 3.8 years ($SD = 5.3$). On average, the maritime crewmembers reported their average tour length to be 66.7 days ($SD = 37.7$), and they were 4.7 days off ($SD = 31.6$) following the tour. Rest periods were 33.5

days on average into their current tour.

The watch schedule was 0600-1800 hours.

The "first" watch schedule was 0600-1800 hours.

The "second" watch schedule was 1800-0600 hours.

The "third" watch schedule was 0600-1800 hours.

The "fourth" watch schedule was 1800-0600 hours.

Methods

Participants

Data were collected from a sample of convenience of one hundred-sixteen crewmembers in the maritime industry. The crewmembers came from eight commercial maritime crews. The crews operated cargo ships that exceed 82 feet in length. Most ships were approximately 180,000 deadweight tons. Six of the ships performed oil transport. Two ships were container ships that carry miscellaneous materials (i.e., cars, food, hazardous material, etc.). Most of the ships would arrive at a port every five to nine days. This would vary depending on weather, schedule changes, and mechanical difficulties. A typical crew comprised of 24 to 28 crewmembers per vessel. Participation in data collection was voluntary. For most cases, 75% of the population was sampled. Data were collected between March 1995 and April 1996 from different people on several ships. None of the ships crossed the International Date Line.

The maritime crewmembers' mean age was 41.4 (SD=11.2), their mean tenure in their profession was 15.9 years (SD=10.9), and their mean tenure at their current watch was 3.8 years (SD=5.3). On average, the maritime crewmembers reported their average tour length to be 86.7 days (SD = 37.7), with 63.7 days off (SD = 31.6) following the tour. Responses were taken 33.5 days (SD = 26.1) on average into their present tour. Typically, 53.0% of the crewmembers' watch schedule was "4-on/8-off", 29.6% is "12-on/12-off", and 17.4% report "Other" watch schedules. Of the crewmembers studied, 47.3% started their first watch before 0600 hours. Mean on-duty sleep length in a 24-hour period was reported as 443.3 minutes per day (SD = 89.3), and off-duty sleep length was reported as 475.7 minutes per day (SD = 104.0).

Materials

The Background Information Inventory (BII) was distributed to the commercial maritime crewmembers. The BII is a 28-page, 119-item inventory containing questions regarding shiftwork, sleep/wake cycles, and opinions of watch scheduling. The BII contained items "from three sources: U.S. Navy research on the operational consequences of sleep deprivation, private sector and NIOSH research conducted by [a] contracted consulting firm on industrial workers employed on shiftwork and other unusual work schedules, and from the Standard Shiftwork Index developed by overseas investigators for European Community studies" (Carvalhais, Tepas, & Paley, 1994). The inventory included sleep problem items that were used to form a composite scale which were related to sleep quality. This measure of sleep quality was provisionally labeled the Sleep Disturbance Scale (Howarth et al., 1999). Neuroticism, extraversion, flexibility of sleeping habits, and chronic fatigue were measured using scales included in the Standard Shiftwork Index (Barton et al., 1995). Specifically, the Neuroticism and Extraversion sub-scales of the Eysenck Personality Inventory (Eysenck, 1964), the Flexibility/Rigidity of Sleeping Habits Scale (Folkard, Monk, & Loban, 1979), and the Chronic Fatigue Scale (Barton et al., 1995) were included in the inventory. The inventory also contained miscellaneous questions about satisfaction with and feelings of sleep and watch schedules.

The Sleep Disturbance Scale is a self-report measure designed to assess quality of sleep (Howarth et al., 1999). The higher the Sleep Disturbance Scale score, the lower the quality of sleep the respondent is reporting. Responses to each of the questions (See Table 5 in Appendix B for a listing of the items) were summed to create a scale score.

The Eysenck Personality Inventory (Eysenck & Eysenck, 1964) is designed to measure personality factors on two different scales. The first is the

Extraversion scale. This scale is said to measure how outgoing and excited a person is (See Table 6 in Appendix B for a listing of the items). Extraverts describe themselves as seeking more frequent and greater amounts of stimulation than introverts (Cooper & Payne, 1967). The second scale is the Neuroticism scale. This scale is said to measure a person's mood state (See Table 7 in Appendix B for a listing of the items). Questions within each of the Eysenck scales were summed to create a score for the scale.

The Flexibility/Rigidity of Sleeping Habits Scale (Folkard et al., 1993; Barton et al., 1995) is said to measure how flexible someone is in terms of changing his or her sleep schedule. The higher the score on this scale, the more flexible a person is in terms of variations in the frequency and duration of their sleep. The scale comes from the Standard Shiftwork Index (Barton et al., 1995). Responses to these questions (See Table 8 in Appendix B for a listing of the items) were summed to create an overall score.

The Chronic Fatigue Scale is part of the Standard Shiftwork Index (Barton et al., 1995). Chronic Fatigue is defined as, "a general tiredness and lack of energy irrespective of whether an individual has not had enough sleep or has been working hard, which persists even on rest days and holidays" (Verhaegen et al., 1981). Positively phrased questions (See Table 9 in Appendix B for a listing of the items) were reverse scored and all questions responses were summed to create a scale score.

Five other questions were asked in the inventory that could be related to sleepiness. "About how often do you feel tired at work?" and "About how often do you feel sleepy at work?" were both assessed on a six-point scale, with 1 being "Never", 3 being "Once or Twice a Month", and 6 being "About Every Day". "Do you feel fatigued during your watch/working period?" was assessed dichotomously, with a "Yes" or a "No" as the possible responses. "How would

you rate your overall physical fitness level?" was assessed on a four-point scale, with 1 being "Poor" and 4 being "Excellent". "How often do you experience seasickness?" was assessed on a 6-point scale, with 1 being "Never", 3 being "Once or Twice a Month", and 6 being "About Every Day". On Duty Sleep Time was calculated from the self-reported average time they went to bed and average time they woke up. Most crewmembers reported two or more sleep periods in a 24-hour period, probably because a majority of the crewmembers worked 4-on/8-off shift schedules. Therefore, Sleep Time was the sum of the sleep times for all sleep periods for each crewmember.

Hypotheses

Eight hypotheses were tested in this study. Pearson product-moment correlations were used to test these hypotheses. The first three hypotheses are designed to test if the measurement of sleep quality correlates with the personality measures of extraversion and neuroticism, the measure of sleeping habit flexibility, and the measure of chronic fatigue. Hypothesis 1a tests the prediction that the Sleep Disturbance Scale will have a statistically significant, negative correlation (Pearson $r < -.3$) with the Extraversion sub-scale of the Eysenck Personality Inventory. The Extraversion sub-scale is designed to measure how outgoing and excited a person is. If this relationship is statistically significant, it will indicate convergent validity between the Sleep Disturbance Scale and the Extraversion sub-scale of the Eysenck Personality Inventory. Hypothesis 1b tests the prediction that the Sleep Disturbance Scale has a statistically significant, positive correlation (Pearson $r > .3$) with the Neuroticism sub-scale of the Eysenck Personality Inventory. The Neuroticism sub-scale of the Eysenck Personality Inventory is designed to measure the arousal level and mood state of an individual. If this relationship is statistically significant, it

indicates evidence for the convergent validity between the Sleep Disturbance Scale and the Neuroticism sub-scale of the Eysenck Personality Inventory. It will also reinforce Åkerstedt and Torsvall's (1981) findings of a significant, positive relationship between the Neuroticism sub-scale and increased sleep complaints.

Hypothesis 2 tests the prediction that sleep quality has a statistically significant, negative correlation (Pearson $r < -.3$) with sleeping habit flexibility. The Flexibility/Rigidity of Sleeping Habits Scale ascertains the flexibility an individual has to changing their normal sleep schedule due to work. Higher scores on this scale are indications of greater flexibility with sleeping habits. If this relationship is statistically significant, it indicates evidence for the convergent validity between the Sleep Disturbance Scale and the Flexibility/Rigidity of Sleeping Habits Scale.

Hypothesis 3 tests the prediction that sleep quality has a statistically significant, positive correlation (Pearson $r > .3$) with chronic fatigue. The Chronic Fatigue Scale is a measure of the general tiredness and lack of energy irrespective of whether an individual has not had enough sleep or has been working hard. Higher scores on this scale are indications of greater feelings of chronic fatigue. If this relationship is found statistically significant, it indicates evidence for the convergent validity between the Sleep Disturbance Scale and the Chronic Fatigue Scale.

The next five hypotheses of this research are intended to test the relationships of sleep quality with other related measures of sleep assessed in the inventory. They test whether the Sleep Disturbance Scale significantly correlates with individual questions regarding sleepiness and fatigue. Hypothesis 4a predicts a statistically significant, positive correlation between the Sleep Disturbance Scale and the inventory item "About how often do you feel tired at work?" (Pearson $r > .3$). Hypothesis 4b predicts a statistically significant, positive

correlation between the Sleep Disturbance Scale and the inventory item "About how often do you feel sleepy at work?" (Pearson $r > .3$). Hypothesis 5 predicts a statistically significant, positive correlation between the Sleep Disturbance Scale and the inventory item: "Do you feel fatigued during your watch period?" (Pearson $r > .3$). Significant correlations between these inventory items and the Sleep Disturbance Scale is a further indication of convergent validity.

Sleep disturbance as measured by the Sleep Disturbance Scale is not designed to measure physiological disturbances. Hypothesis 6 predicts a low (Pearson $r < .1$) correlation between the Sleep Disturbance Scale and the inventory item: "How would you rate your overall physical fitness level?" (Pearson $r > .3$). Hypothesis 7 predicts a low (Pearson $r < .1$) correlation between Sleep Disturbance Scale and the inventory item: "How often do you experience sea sickness?". Low correlations between these inventory items and the Sleep Disturbance Scale indicates evidence for the discriminant validity of the scale.

Hypothesis 8 predicts that sleep quality, as measured by the Sleep Disturbance Scale, significantly correlates with self-reported on-duty sleep length (Pearson $r > .3$). The alternative hypothesis will show a criterion relation between the scale and actual sleep length.

Results

The means and standard deviations of the Sleep Disturbance Scale, the scales from the Standard Shiftwork Index, and the individual questions regarding sleep behavior are shown in Table 1 below. The frequencies of the item responses in these scales can be found in Appendix B. The means for negatively oriented scales, which include the Sleep Disturbance, the Extraversion, and the Chronic Fatigue Scales, are all at least one standard deviation below the scale midpoint.

Table 1. Descriptive Statistics for Scales and Items

	<u>N</u>	<u>Actual</u> <u>Min</u>	<u>Actual</u> <u>Max</u>	<u>Mean</u>	<u>SD</u>	<u>Possible</u> <u>Range</u>
Sleep Disturbance Scale	118	4	15	7.10	2.17	4 to 16
Eysenck Extraversion Scale	118	8	24	17.21	3.35	6 to 24
Eysenck Neuroticism Scale	118	6	16	9.96	2.80	6 to 24
Flexibility/Rigidity of Sleeping Habits	118	26	70	47.28	8.47	14 to 70
Chronic Fatigue Scale	118	10	35	19.38	5.86	10 to 50
About how often do you feel tired at work?	118	1	6	4.06	1.35	1 to 6
About how often do you feel sleepy at work?	118	1	6	3.66	1.44	1 to 6
Do you feel fatigued during your watch/woring period?	118	0	1	.34	.47	0 to 1
How would you rate your overall physical fitness level?	118	1	4	3.14	.63	1 to 4
How often do you experience sea sickness?	118	1	3	1.18	.43	1 to 6
On Duty Sleep Length (minutes)	118	255.0	660.0	449.0	68.4	n/a

Because the standard deviations of the items in the Sleep Disturbance Scale (See Table 10 in Appendix B) are relatively similar, a sum of the scores of the four items will be used as factor scores for the rest of this analysis (Comrey & Lee, 1992). The Confirmatory Factor Analysis of the questions contained in the Sleep Disturbance Scale was reported previously in Howarth, et al. (1999). The internal consistency for the Sleep Disturbance Scale was measured using Cronbach's alpha (Cronbach, 1951), and is shown Table 2. The Cronbach's alpha for the Sleep Disturbance Scale is .78, indicating good internal consistency.

The Standard Shiftwork Index scales used in this analysis were presented previously (Barton et al., 1995). Histograms for the responses to the Standard Shiftwork Index scales in this sample can be found in Appendix B. Since this is a new population, the internal consistency of each of the scales used in this particular sample was calculated. Cronbach's alphas were calculated to measure the reliability for each of the scales in this analysis. These are presented Table 2, with a more detailed listing of these analyses in Table 12 through Table 17 in Appendix B. As can be seen, all Cronbach alphas are greater than .69, indicating good internal consistency of these scales in this sample.

Table 2. Cronbach's Alphas for Scales

<u>Scale</u>	<u>Cronbach's alpha</u>
Sleep Disturbance Scale	.78
Eysenck Extraversion Scale	.76
Eysenck Neuroticism Scale	.69
Flexibility/Rigidity of Sleeping Habits	.83
Chronic Fatigue Scale	.84

Table 3 shows frequencies for the non-scale questions used in this analysis. As is shown in these tables, 68% of the crewmembers report they feel tired at work at least once a week, and 57% report they feel sleepy at work at least once a week. One-third of the crewmembers report they feel fatigued during their watch period, but only one crewmember responded that their physical fitness level was less than fair. In terms of seasickness, 84% reported that they never got seasick. On average, crewmembers reported sleeping 7.5 hours in a 24-hour period. The mean sleep time reported was 449.0 minutes ($N = 116$, $SD = 69.0$). Minimum sleep time was 255 minutes, and the maximum sleep time was 660.0 minutes.

Table 3. Frequencies of Inventory Items

<u>Question</u>	<u>Frequency</u>	<u>Percent</u>
About how often do you feel tired at work?		
Never	6	5.1
Less than once a month	10	8.5
Once or twice a month	21	17.9
Once a week	31	26.5
Two or three times a week	32	27.4
About every day	17	14.5
Total	117	100.0
About how often do you feel sleepy at work?		
Never	13	11.4
Less than once a month	13	11.4
Once or twice a month	23	20.2
Once a week	24	21.1
Two or three times a week	33	28.9
About every day	8	7.0
Total	114	100.0
Do you feel fatigued during your watch/working period?		
No	77	66.4
Yes	39	33.6
Total	116	100.0
How would you rate your overall physical fitness level?		
Poor	1	.9
Fair	13	11.2
Good	71	61.2
Excellent	31	26.7
Total	116	100.0

Question**Frequency Percent****How often do you experience seasickness?**

Never	98	83.8
Less than once a month	17	14.5
Once or twice a month	2	1.7
Total	117	100.0

The hypotheses were tested using Pearson product-moment correlation coefficients between the Sleep Disturbance Scale and the various individual difference scales, specific questions taken from the inventory, and total sleep times. A matrix of these correlation coefficients is presented in Table 4 below. As is indicated in this table, there were several statistically significant correlations. Hypothesis 1a was that the Sleep Disturbance Scale would correlate negatively with the Eysenck Extraversion sub-scale. The correlation between the Sleep Disturbance Scale and the Eysenck Extraversion sub-scale was not statistically significant ($P = .553$, $N = 118$). For Hypothesis 1b, the correlation between the Sleep Disturbance Scale and the Eysenck Neuroticism sub-scale was .382 ($P < .001$, $N = 118$). Hypothesis 2 predicted that the Sleep Disturbance Scale would correlate negatively with the Flexibility/Rigidity of Sleep Habits Scale. A correlation of $-.387$ ($P < .001$, $N = 118$) was found. Hypothesis 3 predicted a positive correlation between the Sleep Disturbance Scale and the Chronic Fatigue Scale. A correlation of $.170$ ($P = .066$, $N = 118$) was found. Hypothesis 4a predicted a positive correlation between the Sleep Disturbance Scale and the inventory item "About how often do you feel tired at work?" A correlation in the hypothesized direction of $.411$ ($P < .001$, $N = 118$) was found. For the inventory item "About how often do you feel sleepy at work?" a correlation of $.369$ ($P < .001$, $N = 118$) was found for Hypothesis 4b. Hypothesis 5 predicted a positive correlation between the Sleep Disturbance Scale and the

inventory item "Do you feel fatigued during your watch period?" a correlation in the hypothesized direction of .229 ($P = .013$, $N = 118$) was found.

Hypothesis 6 predicted a low correlation between the Sleep Disturbance Scale and the inventory item "How would you rate your overall physical fitness level?" A correlation of -.090 ($P = .331$, $N = 118$) was found. Hypothesis 7 also predicted a low correlation between the Sleep Disturbance Scale and the inventory item "How often do you experience seasickness?" A correlation of .130 ($P = .159$, $N = 118$) was found. Hypothesis 8 predicted a negative correlation between the Sleep Disturbance Scale and On Duty Sleep Length. A correlation of -.065 ($P = .484$, $N = 118$) was found.

Table 4. Correlations of Scale Scores

	1	2	3	4	5	6	7	8	9	10
1. Sleep Disturbance Scale - On Duty										
2. Eysenck Extraversion Scale	-.055									
3. Eysenck Neuroticism Scale	.382*	-.020								
4. Flexibility/Rigidity of Sleeping Habits	-.387*	.123	-.258*							
5. Chronic Fatigue Scale	.170	-.290*	.283*	-.074						
6. About how often do you feel tired at work?	.411*	-.092	.478*	-.228*	.435*					
7. About how often do you feel sleepy at work?	.369*	-.103	.384*	-.291*	.365*	.745*				
8. Do you feel fatigued during watch/working period? ^a	.229*	-.132	.381*	-.232*	.278*	.412*	.339*			
9. How would you rate your overall physical fitness level?	-.090	.146	-.101	.069	-.265*	-.129	-.020	.083		
10. How often do you experience sea sickness?	.130	-.077	.143	-.089	.131	.150	.183*	.095	-.121	
11. On Duty Sleep Length	-.065	.115	.062	-.013	-.050	-.144	-.139	-.097	-.131	-.008

* Correlation is statistically significant at the 0.05 level (2-tailed).

^a Spearman's rho correlation coefficient

Discussion

The hypotheses assessed the relationship between the sleep quality and individual difference measures of neuroticism, extraversion, flexibility, and chronic fatigue, as well as several individual measures of being tired, being sleepy, physical fitness, seasickness, and sleep length. Hypothesis 1a stated that the Sleep Disturbance Scale would correlate negatively with the Eysenck Extraversion sub-scale. The correlation between the Sleep Disturbance Scale and the Eysenck Extraversion sub-scale was of a statistically insignificant size.

Eysenck (1964) suggested that introverts differ from extraverts in their arousal threshold. Eysenck (1964) states that extraverted behavior is a function of persistently low cortical arousal and introverted behavior is a function of persistently high cortical arousal. Therefore, introverts require less stimulus input to set their arousal system into an active state. If this is the case, then introverts are more likely to respond to zeitgebers (time givers). They are, therefore, more likely to be aroused in the morning than extraverts (Blake & Corcoran, 1972).

Costa et al. (1989) state that the circadian phase position of biological rhythms and their adjustment during the night shift period seem to be related to various personality and behavioral characteristics. Therefore, these characteristics may be one of the determinants of tolerance to shiftwork. Colquhoun and Folkard (1978) found that "neurotic extraverts" show the greatest degrees of adjustment of body temperature to phase-shifts, while Nachreiner (1975) found that introverts are less tolerant of shiftwork than extraverts. In this sample, however, no support was found for a correlation between extraversion and sleep disturbance.

Hypothesis 1b stated that the Sleep Disturbance Scale would correlate positively with the Eysenck Neuroticism sub-scale. A moderate, positive correlation between these measures supported this part of hypothesis one. Sleep quality is related to negative mood levels in this sample. The lower an individual's mood level, the more sleep disturbance they would report.

Kaliterna, Vidacek, Prizmic, and Radosevic-Vidacek (1995) found that individual difference measures, such as neuroticism and subjective health complaints, are closely related when examined concurrently. When they are measured after a few years of exposure to shiftwork, however, they are not related at all. Contrary to Rutenfranz's (1982) conclusions, Kaliterna et al. (1995) conclude that individual difference measures such as neuroticism are *not* good predictors of health problems as a result of shiftwork. They argue against selection of "healthy" workers based on personality or behavioral characteristics since the predictive validity of these measures was poor. The relationship between neuroticism and sleep quality found in this study replicates Parkes' (1994) findings of a negative relationship of neuroticism and sleep quality.

Hypothesis 2 predicted that the Sleep Disturbance Scale would correlate negatively with the Flexibility/Rigidity of Sleep Habits Scale. A correlation in the hypothesized direction and size of was found, providing support for the second hypothesis. The less flexible crewmembers were in their sleeping habits, the more sleep disturbance they experienced.

Hypothesis 3 predicted a positive correlation between the Sleep Disturbance Scale and the Chronic Fatigue Scale. The size of the correlation coefficient was not as large as was hypothesized, so no support was found for a correlation between chronic fatigue due to shiftwork and sleep disturbance.

Hypothesis 4a predicted a positive correlation between the Sleep Disturbance Scale and the inventory item "About how often do you feel tired at

work?" A statistically significant correlation was found between these variables, so support was found for Hypothesis 4a. The more sleep disturbance crewmembers reported, the more tired they reported feeling while on-duty. Similarly, Hypothesis 4b predicted a positive correlation between the Sleep Disturbance Scale and the inventory item "About how often do you feel sleepy at work?" A statistically significant correlation was found between these variables, so support was found for Hypothesis 4b. The more sleep disturbance crewmembers reported, the sleepier they reported feeling while on-duty. This is similar to and supports Åkerstedt's (1995) findings that deficits in alertness are related to interference with sleep. Reviewing Table 4 in the Results section reveals a strong, positive correlation ($r = .745$, $N = 118$, $P < .001$) between the reported frequency of feeling tired at work and feeling sleepy at work.

Hypothesis 5 predicted a positive correlation between the Sleep Disturbance Scale and the inventory item "Do you feel fatigued during your watch period?" A statistically significant correlation was found, but not of the hypothesized size. There was minimal support for hypothesis five. The more sleep disturbance the crewmembers had, the more fatigue they reported on their watch, but not to the magnitude expected.

Hypothesis 6 predicted a low correlation between the Sleep Disturbance Scale and the inventory item "How would you rate your overall physical fitness level?" This size of this correlation was low, so support was found for the sixth hypothesis. The Sleep Disturbance Scale was not intended to measure health related aspects and their effects on sleep behavior. However, Härmä, Ilmarinen and Knauth (1988) found physical fitness to explain the variations of sleep quality in a sample of female nurses.

Hypothesis 7 predicted a low correlation between the Sleep Disturbance Scale and the inventory item "How often do you experience seasickness?" This

correlation was of an insignificant magnitude, but was not as low as hypothesized. Therefore, no support was found for the seventh hypothesis.

The eighth hypothesis predicted a negative correlation between the Sleep Disturbance Scale and On Duty Sleep Length. This correlation was not of the predicted size, so support was not found for the eighth hypothesis. There was no support for the hypothesis that the lower a crewmembers sleep quality is, the longer they sleep. In their analysis of 1078 surveys from industrial shiftworkers, Mahan, Carvalhais and Queen (1990) also failed to find a relationship between reductions in sleep length and sleep quality.

Knauth, Plett, Condon, DeVol, Fletcher, Eickhoff, and Ottmann (1986) argue that watchkeeping on board sea going ships is different from shiftwork on land. Social "zeitgebers" (time givers) and social problems due to shiftwork are different. Also, in maritime work, upwards of 6 months of continuous work are combined with time-zone transitions. This makes maritime work quite different from factory shiftwork, for example. Knauth et al. (1986) studied the effects of these conditions on 4-on, 8-off watchkeeping system of sea-going ships. They showed that the subjective sleep quality of people doing this kind of watchkeeping does not differ based on the time when the crewmembers did their watch. Their data, as well as the Howarth et al. (1999) data, did show that the watchkeepers had shortened sleep periods on the typical watchkeeping system. Crewmembers experienced a reduction in sleep length, but without a concomitant decrease in sleep quality.

The correlations between the Sleep Disturbance Scale (sleep quality) and the Eysenck Neuroticism (neuroticism) and Flexibility/Rigidity of Sleeping Habits (sleep flexibility) scales are moderate, accounting for about 15% of the variance in each variable. Correlations between the Sleep Disturbance Scale and the Eysenck Extraversion (extraversion) and the Chronic Fatigue scales were not

statistically significant. This provides some evidence of convergent validity of the Sleep Disturbance Scale, because it is related to other measures of sleep habits. The Sleep Disturbance Scale also showed a statistically insignificant relationship to measures of physical fitness and seasickness, which could be measures of fatigue. This provides evidence of the discriminant validity of the Sleep Disturbance Scale, since sleep quality is not directly related to chronic fatigue.

Sleep Disturbance Scale Responses

The Principal Components Analysis of the questions included in the Sleep Disturbance Scale in this sample showed strong loading on a sleep quality component. These four items, "difficulty falling asleep", "difficulty staying asleep", "waking up during sleep", and "difficulty waking up", have been shown to load well in the same manner in another sample of Coast Guard Search and Rescue crews (Howarth et al., 1999). This replication of findings lends greater confidence in using these questions as a scale representative of sleep quality.

As was seen in Table 1 and in the frequencies Tables in Appendix B, the commercial maritime crew in this study responded less than "2" on average for most of the questions in the Sleep Disturbance Scale. In other words, commercial maritime crew responded "Not at all" to "A little" to the questions regarding difficulty falling asleep, staying asleep, waking up during sleep, and difficulty waking up in the morning. About 10% of the commercial maritime crew responded "Quite a bit" or "Almost always" to all but the question about being woken up during sleep. For this question, 25% responded "Quite a bit" or "Almost always". This makes sense, however, in a sample like this. Being on a boat at sea and making scheduled stops at different times of the day, while working 4-on/8-off schedules, can have a tendency to wake people up. The crewmembers reliably used the scale as a measure sleep quality, but they did

not use the full range of the scale. On a scale that has a range of scores from 4 to 16, which would have a theoretical mean of 10, these crewmembers scored a 7 ± 2 on average. This means the crewmembers sleep quality was not very compromised. Comparing the scale to other measures of sleep behavior and related individual difference variables assessed the validity of the scale.

The frequencies for the responses to the scales included from the Standard Shiftwork Index in this analysis also lean towards not having sleep or mood problems. One way of looking at these responses is that classically, commercial maritime crews are "salt of the sea". In other words the crewmembers are rugged individuals who would never have, or would admit to having, any problems while on duty. This may also be the case for the Physical Fitness and Seasickness questions that were asked. Only one person listed their physical fitness level as less than fair, and more than 83% of the commercial maritime crewmembers reported never getting seasick. The commercial maritime crewmembers may be healthy, or they may not want to report that they have any ailments since that could lead to losing their job. Another interpretation is that these individuals are, on average, not "sick", but a few individuals have true sleep, mood and physiological problems.

Tepas (1982) makes a distinction between endogenous and exogenous approaches to studying shiftwork and sleep related problems. The endogenous approach calls for intervention and research efforts aimed at studying biological rhythms, suggesting that an experienced shiftworker may have qualitatively and quantitatively normal sleep. The exogenous approach aims research at removing or preventing deficits, suggesting that complete adaptation to shiftwork is not possible in a day-oriented society. Tepas concluded that in either view, shiftwork changes the lives of workers. The model theorized by Barton et al.

(1995) and the model in this study show how the temporal effects of sleep disturbance can lead to being sleepy at work and eventually to chronic fatigue.

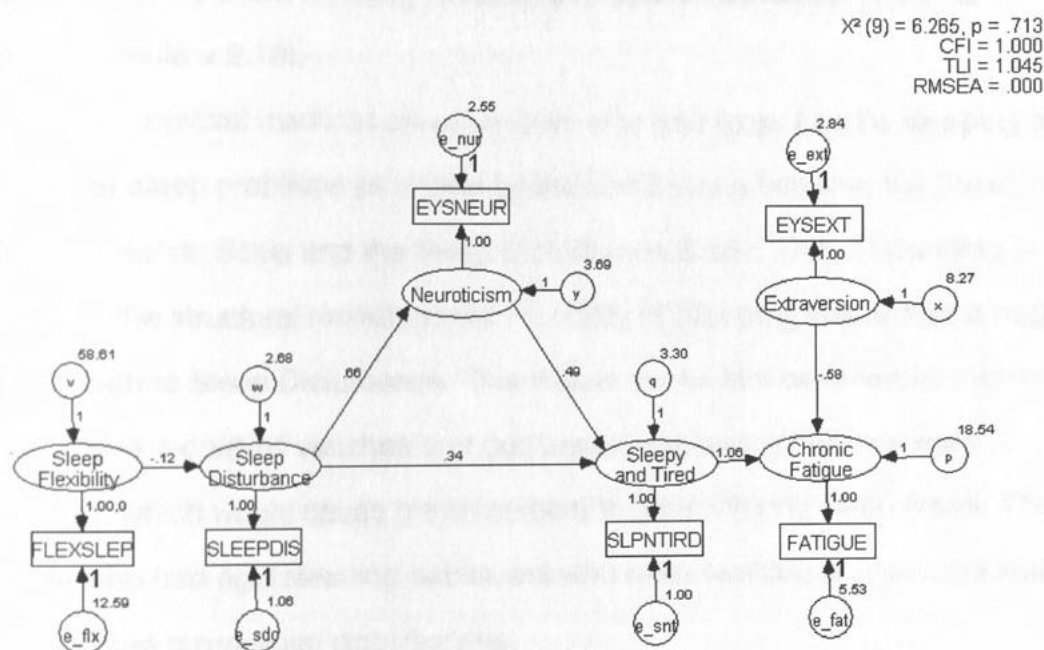
Structural Equation Model of Sleep Scales in this sample

The theoretical model on which the Standard Shiftwork Index (Barton et al., 1995, See Figure 2 in Appendix A) was based, included different shift schedules that would affect the individual difference measures of sleep behavior and quality. These individual differences would in turn cause disturbed sleep. This would result in acute effects on mood, which would lead to chronic effects. Barton et al., (1995, p.25) did mention however, that, "it should be noted that the amount of variance explained was in fact small, and therefore firm conclusions cannot be drawn. What the present results offer is some indication of the importance of the relationship between sleep duration, sleep quality, and health, the complexity of which could be explained in future research."

The correlations in Table 4 are similar in direction and magnitude as the model proposed by Barton et al. (1995). The data for this sample were analyzed by a structural equation model with the specifications established by Barton's theoretical model. All of the variables in the correlation table were entered as scale scores since the sample size was not large enough to account for the variance caused by the sheer number of indicators of all of the scales. The questions of "About how often do you feel sleepy?" and "About how often do you feel tired?" were combined since they were very similar questions and their correlation was high ($r = .745$, $p < .001$). Figure 1 shows the results of the structural equation model. All of the single indicator variables as well as sleep time had insignificant standardized path coefficients in the initial model, so were dropped. The question regarding fatigue during the watch period also had

insignificant standardized path coefficients, possibly due to the fact that the response format to the question was dichotomous.

Figure 1. Structural Model of Sleep Scales in this sample



This model fits the commercial maritime crew data well ($\chi^2(9) = 6.265$). The Comparative Fit Index (CFI) and the Tucker-Lewis Index (TLI) were both 1, and the Root Mean Square Error of Approximation (RMSEA) was .000. All of the paths in this model were statistically significant ($P < .05$). Lower Flexibility of Sleeping Habits caused Sleep Disturbance. A higher Sleep Disturbance caused more Neuroticism and a person to be more Sleepy/Tired. Neuroticism in turn caused a person to be Sleepy/Tired as well. Being more Sleepy/Tired caused more Chronic Fatigue, which was also caused by a lower score in Extraversion.

The path from Sleep Disturbance to Sleepy/Tired was tested for mediation by Neuroticism using Baron and Kenny's (1986) method. A χ^2 difference test

between the model with a deleted path between Sleep Disturbance and Sleepy/Tired measure was compared to a model including the path. The Chi-Square difference test showed mediation ($\chi^2_{diff}(1) = 3.761, P < .05$). A test of mediation using Sobel's (1982) formula also reached statistical significance (Sobel formula = 2.70).

Commercial maritime crewmembers who had more flexible sleeping habits had fewer sleep problems as shown by the correlations between the Flexibility of Sleeping Habits Scale and the Sleep Disturbance Scale. This relationship is also shown in the structural model, where Flexibility of Sleeping Habits had a negative causal path to Sleep Disturbance. This makes sense in a commercial maritime crew where 4-on/8-off watches and odd breaks in sleep schedules were common, which would cause crewmembers to have varying sleep times. Those people who had rigid sleeping habits, but who were working on maritime watches encountered more sleep disturbances.

Those commercial maritime crewmembers who reported more sleep disturbances also reported lower mood as reported by the Neuroticism sub-scale of the Eysenck Personality Inventory. Both sleep disturbances and neuroticism were related to being sleepy and tired on the job. The model shows that having disturbed sleep causes a lowered mood state as well as being tired on the job. A mediation analysis of the relationship between Sleep Disturbance and being Sleepy and Tired shows that that relationship is mediated by neuroticism. Having a bad night sleep may cause commercial maritime crewmembers to be sleepy and tired the next day, but this relationship depends upon their mood state.

The relationship between being sleepy and tired at work and chronic fatigue is fairly obvious. The more commercial maritime crewmembers are sleepy and tired on the job, the more long-term fatigue effects they may have. The

causal direction of this can be seen in the model where Sleepy/Tired causes Chronic Fatigue.

Interestingly enough, Barton et al. (1995) discussed how inter-individual differences would affect measures of sleepiness. In her model, Eysenck Neuroticism was shown to be a mediator of the Sleep Disturbance and the Sleepy/Tired relationship. The Eysenck measure of Extraversion showed a different kind of relationship in the current sample of maritime crewmembers. Extraversion was a causal factor of Chronic Fatigue. If commercial maritime crewmembers reported that they were more lively and outgoing, they experienced less chronic fatigue.

This model is similar to that found by Barton et al. (1995). Their model showed the individual difference variable of Flexibility of Sleeping Habits causing another measure of sleep quality that in turn caused Chronic Fatigue. These relationships were moderated by Neuroticism in their model. In the model shown in this study, the individual differences variable Neuroticism played a mediating role with sleep quality and being sleepy and tired on watch.

The model for the commercial maritime crews in this study is different than the Barton et al. (1995) theoretical model in a few aspects, however. First, the Barton et al. model theorizes a shift system effect as part of the model. Not all of commercial maritime crews in this study worked the same shift system. The sample size was not large enough to do an adequate test of the model considering the different watches in this sample. A future study of this model in relation to commercial maritime crewmembers should include a comparison of the model in different watch schedules. This model also differs from the theoretical model in the sample it studied. Barton's sample was mostly female nurses, while this study looked at male commercial crewmembers. The participants in the Barton study worked 10 or more hour workdays, and these

crewmembers did not. Barton's participants were also much younger than these crewmembers. However, since there is some agreement between the model in this sample and Barton's model, and since both the nurses and commercial maritime crewmembers are not representative of all workers, caution should be taken in applying this model to new samples.

Finally, this model differs from the theoretical model in that a different measure of sleep disturbance was used. Since the results of this analysis are similar to the predicted model that uses a different measure of sleep disturbance gives evidence of the construct validity of the Sleep Disturbance Scale.

General Discussion

Maritime workers on ocean-going vessels differ from regular workers in that they carry their "home" with them wherever they go according to Colquhoun (1985). Maritime workers are constantly "living on the job". Unlike factory workers, maritime workers sleep in the place that they work, and their lives depend upon a "factory" that is functioning 24 hours a day. On top of this, their environment is in a continuous state of movement and can be hot and noisy. These conditions can provide for poor sleeping conditions.

Colquhoun (1985) describes the typical watchkeeping schedule of maritime crews. Traditionally, a watch is four hours in length. This practice was established in the days of sailing ships, when the work needing to be done was arduous, and had to be done on an open deck under all kinds of weather conditions. Although maritime crewmembers of today do not need to brave the elements like they used to, the four-hour watch tradition is still maintained. A 24-hour period is divided into six watches; 0000-0400 (middle), 0400-0800 (morning), 0800-1200 (forenoon), 1200-1600 (afternoon), 1600-2000 (dogs), and 2000-0000 (first). Typically, commercial maritime crewmembers are assigned to

pairs of these watches, with two watches off in between, and are therefore called 4-on, 8-off watches. Assignment to these watches is usually based on seniority. These schedules can also lead to sleep problems.

Kleitman (1949) found in his study of submarine watchkeepers that a majority of the participants took two or more sleep periods because of this type of schedule. Fletcher et al. (1986) argue that in the modern Merchant Navy, neither the "close" (where the 4 hour watch periods occur close to each other in time during the day), nor the "shiftworking" system (where 8 hour shifts would rotate throughout a day) is practical. This is because neither of these schedules was designed for vessels where the numbers of crew and watchkeepers have been cut to a minimum. Fletcher proposed a "2/6" system, where two watches, one of two hours, one of eight hours, would be worked in a given day. This would allow for more normal sleep periods.

Fletcher, Colquhoun, Knauth, DeVoul, and Plett (1986, 1988) research the disadvantages of a 4-on/8-off shiftwork scheduling and conclude that sleep would be better under the alternative "2 hour/6 hour" watch schedule. Unfortunately, Cole, Loving, and Kripke (1990) report that there is little to no rigorous data supporting the notion that exposure to shiftwork like this leads to abnormal, disturbed, or disordered workers. Based on a review of shiftwork research, Åkerstedt (1995) summarizes that sleepiness in night work is associated with performance impairments to the point that it explains night-work accident data. How do we measure the effects of these watches on sleep and mood in a maritime crew?

The commercial maritime crewmembers responding to the inventory used in this study were representative of the merchant marine industry. They operated large tanker-type ships, transporting cargo from various ports in northwestern United States. The crewmembers were middle-aged, and, on average, had been

in their profession for a long time. They reported that they went on long tours that lasted about three months on average. Responses to the inventory were taken one month into their tour. More than half of the commercial maritime crewmembers worked a 4-on/8-off watch schedule. Almost half of them started their workday before 0600 hours. They reported sleeping a little more than 7 hours a day when on-duty, but in a majority of cases this sleep time was split into two sleep sessions in a 24-hour period due to their watch schedule. Although not extreme, this type of schedule is not ideal. One would assume that the commercial maritime crew in this sample would experience sleepiness and fatigue as a result of their schedules. Assessing subjective sleep measures in this population should have brought about some extremes for analysis, however it did not in this sample. Most of the commercial maritime crew responded to the Sleep Disturbance Scale below the scale midpoint, indicating that they are not suffering from disturbed sleep, even though a majority of them reported being sleepy and tired on their watches. Minor sleep disturbances can result in being sleepy and tired.

The Howarth, Pratt and Tepas (1999) research was done to, "examine self-reports of sleep duration and sleep problems of maritime crew members employed on shift schedules for continuous operations." (Howarth et al., 1999). The research was not intended to evaluate whether the maritime crewmembers are suffering from a significant sleep disturbance, but simply to establish a measure of sleep quality. The purpose of the present study was to evaluate the validity of this scale.

Howarth et al. (1999) found no differences between on- and off-duty sleep quality as measured by the Sleep Disturbance Scale in two commercial maritime crewmember samples. In this report it was shown that the Sleep Disturbance Scale had no statistically significant relationship with sleep length. Two

interpretations of this are that 1) the Sleep Disturbance Scale is not a valid measure of sleep quality, or 2) in these samples, maritime duty had an insignificant impact on sleep quality. As a result of the statistically significant correlations between the Sleep Disturbance Scale and other measures of sleep behavior and mood, as well as a similar model fit as that of Barton et al. (1995), it is concluded that the second interpretation is true. In this sample of commercial maritime crewmembers, the Sleep Disturbance Scale is a valid measure of sleep quality, and therefore the suggestion made by Howarth et al. (1995) that maritime duty had no effect on sleep quality is accurate.

Suggestions for Future Research

Even though the inventories in this study were collected over a year's time, the data only indicates a snapshot in time of sleep behaviors. Variables like chronic fatigue, as the name suggests, happen over time. In future studies, assessing the longitudinal effects of schedule flexibility, sleep disturbance, the mediating effects of neuroticism, and chronic fatigue on actual performance can lead to a better understanding of the effects of sleep behavior and individual difference measures on the performance of commercial maritime crewmembers.

All of the questions in the inventory that the commercial maritime crewmembers answered were self-report, Likert-scaled items. The correlations between the different measures of sleep behavior and mood may lend evidence to the convergent validity of the Sleep Disturbance Scale, but a more detailed analysis can be done. The correlational method used in this study did not tease apart the difference between method variance and trait variance of the different measures. The correlations between the variables in this study may be solely due to the questionnaire method used in this study, and not due to the latent trait variance the scales are meant to measure. In future studies, a multitrait-

multimethod matrix (Campbell & Fiske, 1959) analysis of these measures would further lend to the convergent and discriminant validity of the Sleep Disturbance Scale.

This study should not lead to the conclusion that this model is applicable to other populations. Different work schedules in different environments can have different effects on sleep quality, which could result in different causal effects of the variables studied here. The Sleep Disturbance Scale, as well as Barton's et al.'s, (1995) model, should be tested in other populations before making generalizations from them.

Conclusion

Problems due to sleep and shiftwork are multifaceted. In the maritime industry, watchkeeping schedules can affect sleep and sleep quality. Sleep quality can be measured using simple subjective measures. The Sleep Disturbance Scale measure of sleep quality has a predicted relationship with neuroticism, sleep schedule flexibility, and being sleepy and tired in this sample of commercial maritime crewmembers. The title "Sleep Disturbance Scale" is in fact the right label for this measure of sleep quality, and the commercial maritime crewmembers do use it as a scale in a restricted way.

A strong note of caution should be taken in reviewing these findings, however. Sleep quality as measured by the Sleep Disturbance Scale in this sample showed just minor variations accounting for a modest amount of the total variance of the scale. Commercial maritime crewmembers as a rule are fairly normal people working an abnormal work schedule. A casual reading of this study, as well as the Barton et al. (1995) study, should *not* leave the reader feeling that commercial maritime crewmembers and nurses are sick people, but

people having normal variance in their sleep quality, possibly due to their shift schedules.

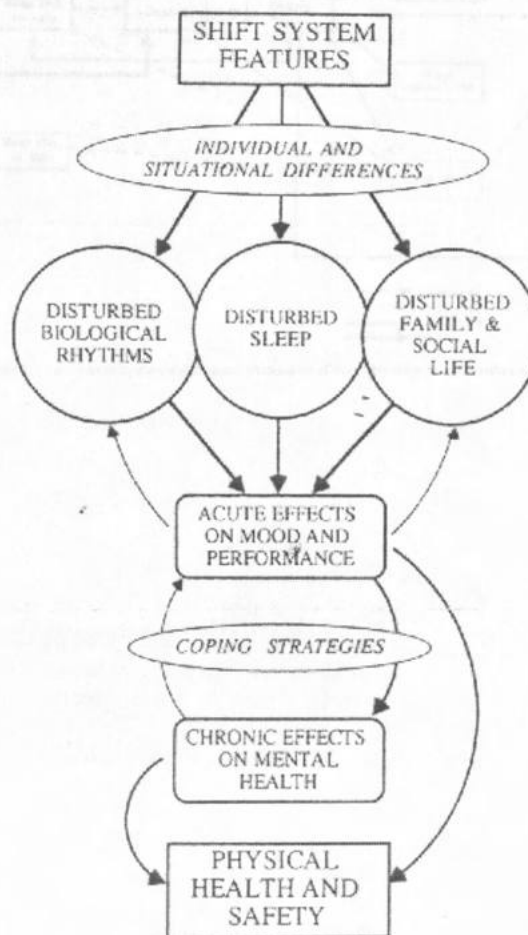
But can the measures used in this study be used to avoid catastrophes like the Valdez? It cannot be said from the results of this study since there are no measures of the effects these scales have on actual performance. The Sleep Disturbance Scale measure of sleep quality, however, does provide us with a promising means to understanding aspects of sleep behavior. These aspects can be used to identify how sleep behavior can affect performance on the job due to the quality of sleep.



Appendix A

The following figure (reprinted from Barton et al. (1995) with permission). The figure shows the theoretical model Barton set out to test using measures from the Standard Shiftwork Index.

Figure 2. Standard Shiftwork Index Theoretical Model from Barton et al., (1995)



The following figure (printed with permission) shows the final model of the relationship between the different scales included in the Standard Shiftwork Index.

Figure 3. Final Barton et al. (1995) Model

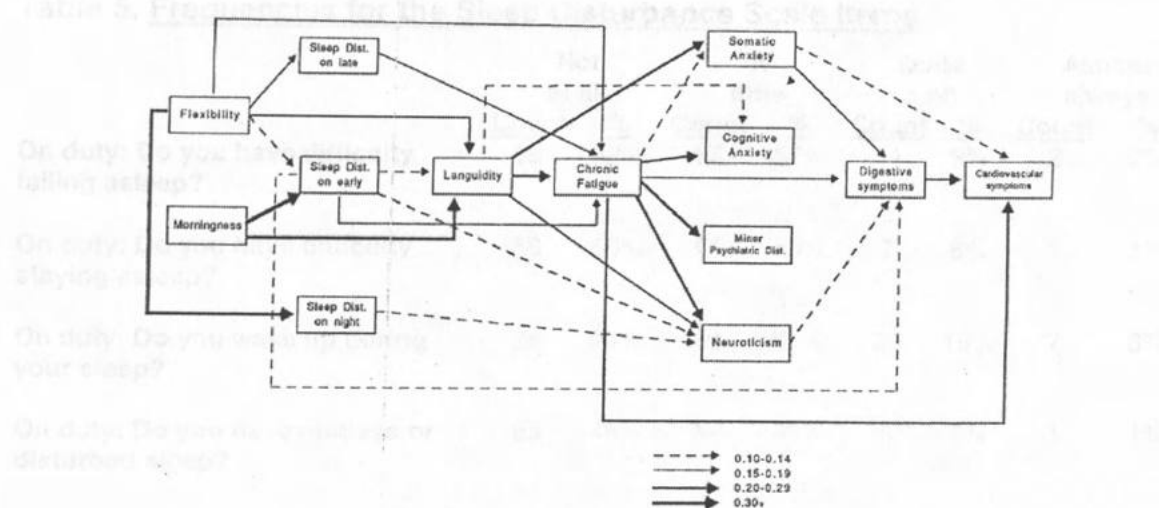


Table 6. Frequencies for Greenick Extraversion Inventory Items

	Never	Seldom	Often	Always
Do you like plenty of excitement and stimulation?	10	9%	37	32%
Are you rather shy?	2	2%	24	21%

Appendix B

The following tables contain the inventory items and their response frequencies in this sample for each of the scales used in this study.

Table 5. Frequencies for the Sleep Disturbance Scale Items

	Not at all		A little		Quite a bit		Almost always	
	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>
On duty: Do you have difficulty falling asleep?	38	32%	67	57%	11	9%	2	2%
On duty: Do you have difficulty staying asleep?	59	50%	51	43%	7	6%	1	1%
On duty: Do you wake up during your sleep?	28	24%	60	51%	23	19%	7	6%
On duty: Do you have restless or disturbed sleep?	53	45%	54	46%	9	8%	1	1%

Table 6. Frequencies for Eysenck Extraversion Inventory Items

	Almost Never		Quite seldom		Quite often		Almost always	
	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>
Do you like plenty of excitement and bustle around you?	10	9%	37	33%	44	40%	20	18%
Are you rather lively?	2	2%	24	21%	67	58%	22	19%
Do you like mixing with people?	2	2%	30	27%	50	44%	31	27%
Would you call yourself happy-go-lucky?	13	11%	24	21%	51	45%	26	23%
Can you let yourself go and enjoy yourself a lot at a lively party?	3	3%	21	18%	44	39%	46	40%
Do you like practical jokes?	16	14%	35	30%	35	30%	29	25%

Table 7. Frequencies for Eysenck Neuroticism Inventory Items

	Almost Never		Quite seldom		Quite often		Almost always	
	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>
Does your mood go up and down?	39	34%	45	39%	27	23%	4	3%
Do you feel just miserable for no good reason?	86	75%	24	21%	4	3%	1	1%
When you get annoyed, do you need someone to talk to?	37	32%	39	34%	32	28%	7	6%
Are you troubled about feelings of guilt?	72	63%	28	24%	11	10%	4	3%
Would you call yourself tense or high strung?	63	55%	38	33%	12	10%	2	2%
Do you suffer from sleeplessness?	66	58%	40	35%	6	5%	1	1%

Table 8. Frequencies for Flexibility/Rigidity of Sleeping Habits Scale Items

	Almost Never		Seldom		Sometimes		Usually		Almost always	
	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>
If you are feeling drowsy can you easily overcome it if you have something to do?	2	2%	1	1%	14	12%	71	62%	27	23%
Do you find it fairly easy to get to sleep whenever you want to?	4	3%	11	9%	26	22%	47	41%	28	24%
Can you miss a sleep period without too much difficulty?	3	3%	13	11%	36	31%	48	41%	17	15%
Do you think you could do a job in the middle of the night as easy as at a more normal time of day?	4	3%	10	9%	25	22%	53	46%	24	21%
Do you find it easy to sleep in the morning if you got to bed very late the previous night?	8	7%	18	16%	26	23%	35	30%	28	24%
Can you keep alert in boring situations?	7	6%	22	19%	31	27%	46	40%	8	7%

	Almost Never		Seldom		Sometimes		Usually		Almost always	
	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>
Do you enjoy working at unusual times of day or night?	7	6%	22	19%	33	29%	33	29%	19	17%
If you have a lot to do, can you stay up late to finish it off without feeling too tired?	3	3%	12	10%	24	21%	53	46%	23	20%
Do you find it as easy to work late at night as earlier in the day?	6	5%	21	18%	28	24%	41	36%	19	17%
Can you easily go to sleep earlier than normal to catch up on lost sleep?	13	11%	13	11%	22	19%	37	32%	30	26%
Do you have no strong preferences as to when you sleep?	16	14%	35	31%	22	20%	28	25%	11	10%
Can you manage with only a few hours sleep for several days in a row without too much difficulty?	12	10%	16	14%	30	26%	40	34%	18	16%
Would you be just as happy to do something in the middle of the night as during the day?	16	14%	31	27%	27	23%	29	25%	13	11%
Do you go to parties or have evenings out with friends if you have to get up early?	28	24%	28	24%	38	33%	14	12%	8	7%

Table 9. Frequencies for Chronic Fatigue Scale Items

	Not at all		A little		Somewhat		More than somewhat		Very much	
	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>
I generally feel I have plenty of energy.	2	2%	4	4%	31	27%	36	32%	41	36%
I usually feel drained.	49	43%	43	38%	13	12%	7	6%	1	1%
I generally feel quite active.	5	4%	5	4%	32	28%	34	30%	39	34%

	Not at all		A little		Somewhat		More than somewhat		Very much	
	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>	<u>Count</u>	<u>%</u>
I feel tired most of the time.	54	48%	41	37%	13	12%	3	3%	1	1%
I usually feel full of vigor.	5	5%	8	7%	45	41%	31	28%	22	20%
I usually feel rather lethargic.	57	53%	34	31%	14	13%	2	2%	1	1%
I generally feel alert.	3	3%	2	2%	26	23%	44	39%	38	34%
I often feel exhausted.	52	46%	40	36%	15	13%	3	3%	2	2%
I usually feel lively.	4	4%	3	3%	34	30%	41	36%	32	28%
I feel weary much of the time.	69	62%	33	29%	6	5%	2	2%	2	2%

The following figures show the histograms of the scales. A normal curve is overlaid onto the figure.

Figure 4. Histogram of Sleep Disturbance Scale

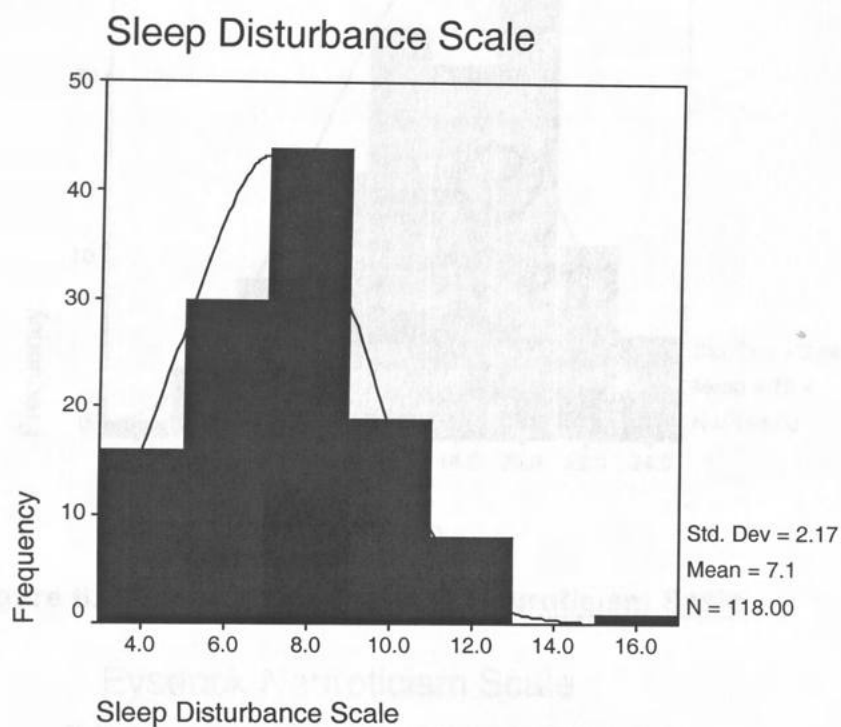


Figure 5. Histogram for Eysenck Extraversion Scale

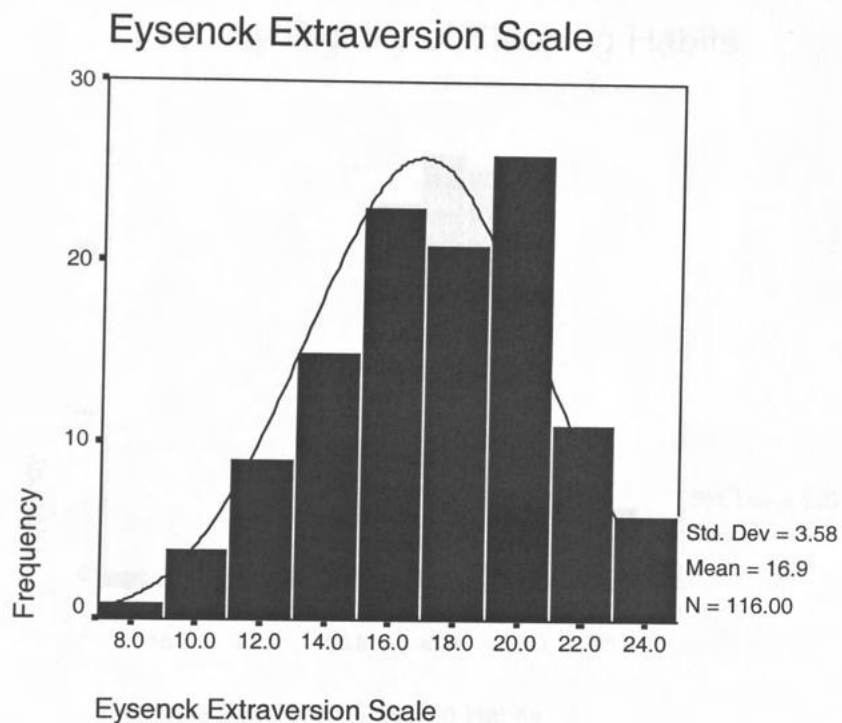


Figure 6. Histogram for Eysenck Neuroticism Scale

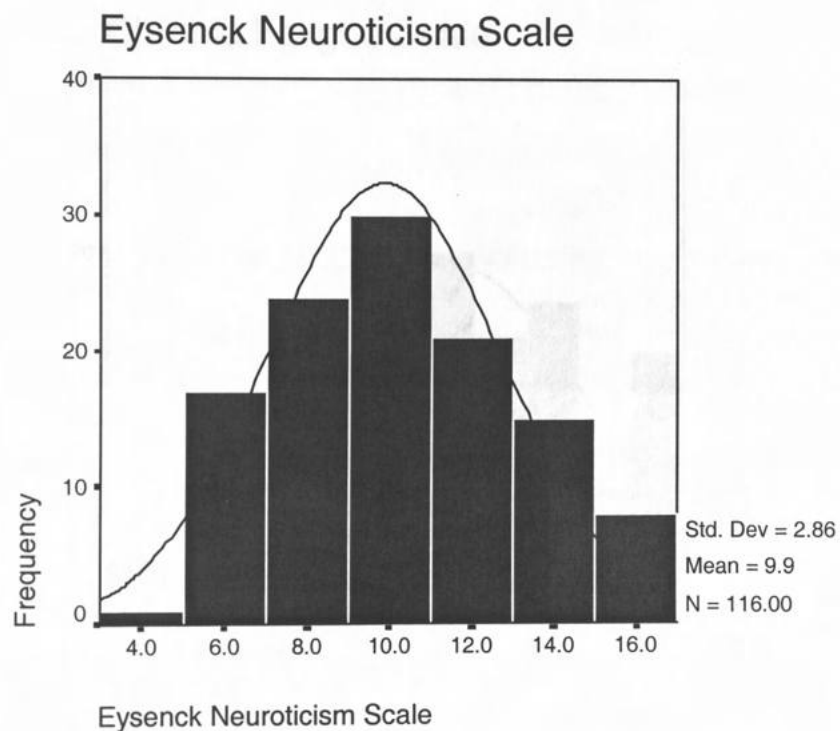
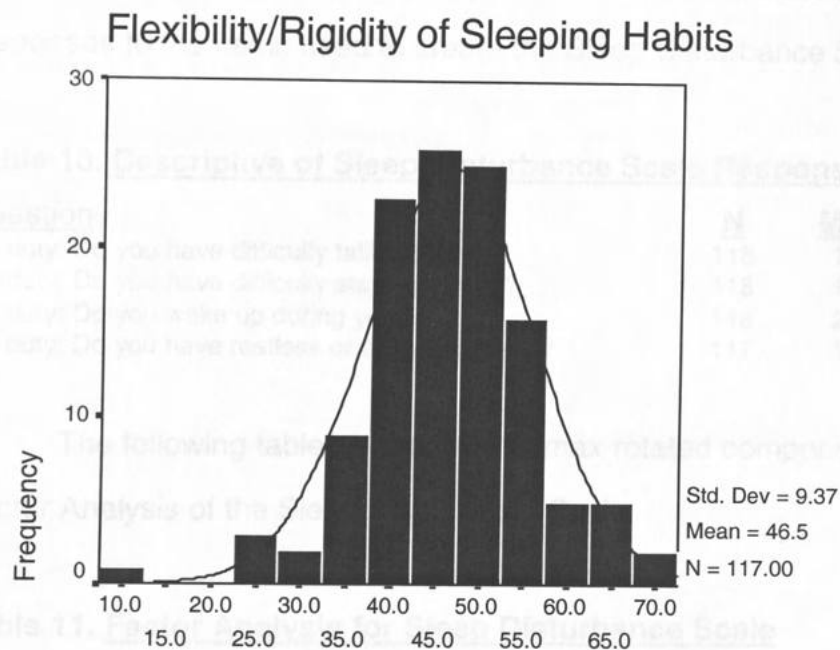
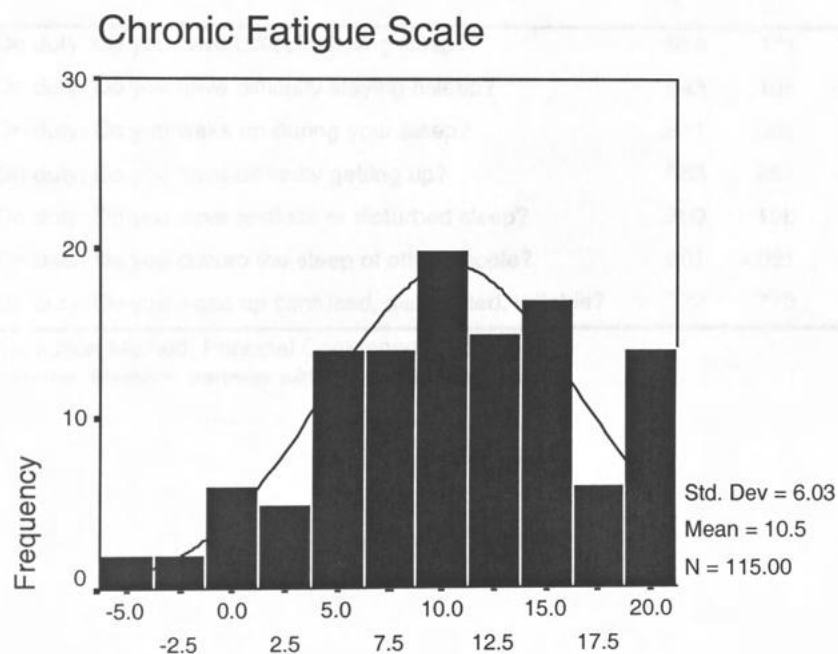


Figure 7. Histogram for Flexibility/Rigidity of Sleeping Habits Scale



Flexibility/Rigidity of Sleeping Habits

Figure 8. Histogram of Chronic Fatigue Scale



Chronic Fatigue Scale

The following table shows the mean and standard deviation of the responses to the items used to create the Sleep Disturbance Scale scores.

Table 10. Descriptive of Sleep Disturbance Scale Responses

<u>Question</u>	<u>N</u>	<u>Mean</u>	<u>SD</u>
On duty: Do you have difficulty falling asleep?	118	1.81	.67
On duty: Do you have difficulty staying asleep?	118	1.58	.65
On duty: Do you wake up during your sleep?	118	2.08	.82
On duty: Do you have restless or disturbed sleep?	117	1.64	.66

The following table shows the Varimax rotated component matrix for the Factor Analysis of the Sleep Disturbance Scale.

Table 11. Factor Analysis for Sleep Disturbance Scale

Rotated Component ^a			
	Component		
	1	2	3
On duty: Do you have difficulty falling sleep?	.594	.371	-.259
On duty: Do you have difficulty staying asleep?	.843	.105	-.122
On duty: Do you wake up during your sleep?	.811	.086	.035
On duty: Do you have difficulty getting up?	.088	.851	-.090
On duty: Do you have restless or disturbed sleep?	.760	.198	.296
On duty: Do you disturb the sleep of other people?	.001	-.001	.957
On duty: Do you wake up confused, disoriented, irritable?	.222	.773	.112

Extraction Method: Principal Component Analysis.

Rotation Method: Varimax with Kaiser Normalization.

a. Rotation converged in 5 iterations.

The following tables show the reliability analyses (Cronbach, 1951) of each of the scales.

Table 12. Reliability - Sleep Disturbance Scale

RELIABILITY ANALYSIS - SCALE (ALPHA)

		Mean	Std Dev	Cases
1.	C58_1	1.8051	.6700	118.0
2.	C59_1	1.5763	.6459	118.0
3.	C60_1	2.0763	.8181	118.0
4.	C62_1	1.6410	.6597	118.0

N of Cases = 118.0

				N of		
Statistics for	Mean	Variance	Std Dev	Variables		
Scale	7.0987	4.6983	2.1676	4		
Item Means	Mean	Minimum	Maximum	Range	Max/Min	Variance
	1.7747	1.5763	2.0763	.5000	1.3172	.0497
Item Variances	Mean	Minimum	Maximum	Range	Max/Min	Variance
	.4927	.4172	.6693	.2521	1.6043	.0140

Reliability Coefficients 4 items

Alpha = .7741

Standardized item alpha = .7778

Table 13. Reliability - Eysenck Extraversion Inventory

R E L I A B I L I T Y A N A L Y S I S - S C A L E (A L P H A)

		Mean	Std Dev	Cases
1.	C122_1	2.6667	.8507	118.0
2.	C124_1	2.9478	.6774	118.0
3.	C126_1	2.9735	.7675	118.0
4.	C128_1	2.7895	.9103	118.0
5.	C130_1	3.1667	.8051	118.0
6.	C132_1	2.6696	.9933	118.0

N of Cases = 118.0

Statistics for	N of			
	Mean	Variance	Std Dev	Variables
Scale	17.2136	11.2032	3.3471	6

Item Means	Mean	Minimum	Maximum	Range	Max/Min	Variance
	2.8689	2.6667	3.1667	.5000	1.1875	.0386

Item Variances	Mean	Minimum	Maximum	Range	Max/Min	Variance
	.7058	.4589	.9867	.5278	2.1503	.0344

Reliability Coefficients 6 items

Alpha = .7464

Standardized item alpha = .7609

Table 14. Reliability - Eysenck Neuroticism Inventory

RELIABILITY ANALYSIS - SCALE (ALPHA)

		Mean	Std Dev	Cases		
1.	C123_1	1.9652	.8365	118.0		
2.	C125_1	1.3043	.5725	118.0		
3.	C127_1	2.0783	.9072	118.0		
4.	C129_1	1.5391	.7984	118.0		
5.	C131_1	1.5913	.7384	118.0		
6.	C133_1	1.4867	.6286	118.0		
N of Cases =		118.0				
N of						
Statistics for	Mean	Variance	Std Dev	Variables		
Scale	9.9650	7.8396	2.7999	6		
Item Means	Mean	Minimum	Maximum	Range	Max/Min	Variance
	1.6608	1.3043	2.0783	.7739	1.5933	.0888
Item Variances	Mean	Minimum	Maximum	Range	Max/Min	Variance
	.5714	.3278	.8230	.4953	2.5111	.0350
Reliability Coefficients		6 items				
Alpha = .6752		Standardized item alpha =		.6882		

Table 15. Reliability - Flexibility/Rigidity of Sleeping Habits Scale

R E L I A B I L I T Y A N A L Y S I S - S C A L E (A L P H A)						
		Mean	Std Dev	Cases		
1.	C140_1	4.0435	.7325	118.0		
2.	C141_1	3.7241	1.0343	118.0		
3.	C142_1	3.5385	.9567	118.0		
4.	C143_1	3.7155	.9941	118.0		
5.	C144_1	3.4957	1.2010	118.0		
6.	C145_1	3.2281	1.0214	118.0		
7.	C146_1	3.3070	1.1257	118.0		
8.	C147_1	3.7043	.9782	118.0		
9.	C148_1	3.4000	1.1079	118.0		
10.	C149_1	3.5043	1.2835	118.0		
11.	C150_1	2.8482	1.1998	118.0		
12.	C151_1	3.3103	1.1869	118.0		
13.	C152_1	2.9310	1.2246	118.0		
14.	C153_1	2.5345	1.1726	118.0		
N of Cases =		118.0				
N of						
Statistics for	Mean	Variance	Std Dev	Variables		
Scale	47.2851	71.8088	8.4740	14		
Item Means	Mean	Minimum	Maximum	Range	Max/Min	Variance
	3.3775	2.5345	4.0435	1.5090	1.5954	.1587
Item Variances	Mean	Minimum	Maximum	Range	Max/Min	Variance
	1.2012	.5366	1.6474	1.1108	3.0701	.0886
Reliability Coefficients		14 items				
Alpha = .8247		Standardized item alpha = .8254				

Table 16. Reliability - Chronic Fatigue Scale

RELIABILITY ANALYSIS - SCALE (ALPHA)

		Mean	Std Dev	Cases
1.	C112_1	2.0351	.9512	118.0
2.	C113_1	1.8319	.9049	118.0
3.	C114_1	2.1565	1.0669	118.0
4.	C115_1	1.7143	.8210	118.0
5.	C116_1	2.4865	1.0031	118.0
6.	C117_1	1.6667	.8060	118.0
7.	C118_1	2.0088	.9198	118.0
8.	C119_1	1.7768	.8839	118.0
9.	C120_1	2.1754	.9718	118.0
10.	C121_1	1.5268	.8055	118.0

N of Cases = 118.0

				N of
Statistics for	Mean	Variance	Std Dev	Variables
Scale	19.3788	34.3872	5.8641	10

Item Means	Mean	Minimum	Maximum	Range	Max/Min	Variance
	1.9379	1.5268	2.4865	.9597	1.6286	.0833

Item Variances	Mean	Minimum	Maximum	Range	Max/Min	Variance
	.8412	.6489	1.1383	.4894	1.7543	.0262

Reliability Coefficients 10 items

Alpha = .8393 Standardized item alpha = .8391

Table 17. Reliability – Sleepy and Tired Items Combined

RELIABILITY ANALYSIS - SCALE (ALPHA)

		Mean	Std Dev	Cases
1.	C107_1	4.0598	1.3543	118.0
2.	C108_1	3.6579	1.4431	118.0

N of Cases = 118.0

		Mean	Variance	Std Dev	N of Variables
Statistics for					
Scale		7.7177	6.8293	2.6133	2

Item Means	Mean	Minimum	Maximum	Range	Max/Min	Variance
	3.8589	3.6579	4.0598	.4019	1.1099	.0808

Item Variances	Mean	Minimum	Maximum	Range	Max/Min	Variance
	1.9583	1.8340	2.0825	.2485	1.1355	.0309

Reliability Coefficients 2 items

Alpha = .8530 Standardized item alpha = .8540

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Reference List

- Åkerstedt, T. (1995). Work hours, sleepiness and the underlying mechanisms. Journal of Sleep Research, 4(Suppl. 2), 15-22.
- Åkerstedt, T., & Torsvall, L. (1981). Shift Work: Shift-dependant well-being and individual differences. Ergonomics, 24(4), 265-273.
- Baron, R. M., & Kenny, D. A. (1986). The moderator-mediator variable distinction in social psychological research: Conceptual, strategic and statistical considerations. Journal of Personality and Social Psychology, 51, 1173-1182.
- Barton, J., Spelten, E., Totterdell, P., Smith, L., & Folkard, S. (1995). Is there an optimum number of night shifts? Relationship between sleep, health and well-being. Work and Stress, 9(2/3), 109-123.
- Barton, J., Spelten, E., Totterdell, P., Smith, L., Folkard, S., & Costa, G. (1995). The Standard Shiftwork Index: a battery of questionnaires for assessing shiftwork-related problems. Work and Stress, 9(1), 4-30.
- Bruusgaard, A. (1969). Shift work as an occupational health problem. Night and Shift Work Stockholm, Sweden: Institute of Occupational Health.
- Campbell, D. and Fiske, D. (1959). Convergent and discriminant validation by the

- multitrait-multimethod matrix. Psychological Bulletin, 56(2), 81-105.
- Carvalhois, A. B., Tepas, D. I., & Paley, M. J. (1994). An Evaluation of a Coast Guard "Live-Aboard" Concept: Can crews adapt to a restricted living and operational environment?. Proceedings of the Human Factors and Ergonomics Society 38th Annual Meeting – 1994.
- Cohen, J. (1992). Statistical Power Analysis for the Behavioral Sciences. 2nd Ed. Hillsdale, NJ: Erlbaum. ISBN: 0-8058-0283-5
- Cole, R. J., Loving, R. T., and Kripke, D. F. (1990). Psychiatric Aspects of Shiftwork. Occupational Medicine: State of the art reviews, 5(2), 301-314.
- Colquhoun, W. P. (1985). Hours of work at sea: watchkeeping schedules, circadian rhythms and efficiency. Ergonomics, 28(4), 637-653.
- Colquhoun, W. P., & Folkard, S. (1978). Personality Differences in Body-temperature Rhythm, and their Relation to its Adjustment to Night Work. Ergonomics, 21(10), 811-817.
- Comrey, A. L., & Lee, H. B. (1992). A first course in factor analysis (2nd ed.). Hillsdale, NJ: Lawrence Erlbaum Associates, Inc.
- Cooper, R., & Payne, R. L. (1967). Extraversion and some aspects of work behavior. Personnel Psychology, 20, 45-57.

- Costa, G., Lievore, F., Casaletti, G., Gaffuri, E., & Folkard, S. (1989). Circadian characteristics influencing interindividual differences in tolerance and adjustment to shiftwork. Ergonomics, 32(4), 373-385.
- Cronbach, L. J. (1951). Coefficient alpha and the internal structure of tests. Psychometrika, 16, 297-334.
- de Vries & de Vries (1989). The Process of Developing Health Complaints: A Longitudinal Study of the Effects of Abnormal, Irregular and Condensed Working Hours. In Costa, G., Cesana, G., Kogi, K. & Wedderburn, A. (Eds.), Shiftwork: Health, Sleep and Performance, (pp 290-296). New York: Peter Lang.
- Eysenck, H. J. (1976). The Measurement of Personality. Baltimore, Maryland: University Park Press.
- Eysenck, S. B. G., & Eysenck, H. J. (1964). An improved short questionnaire for the measurement of extraversion and neuroticism. Life Sciences, 3, 1103-1109.
- Fletcher, N., Colquhoun, W. P., Knauth, P., DeVol, D., & Plett, R. (1988). Work at sea: a study of sleep, and of circadian rhythms in physiological and psychological functions, in watchkeepers on merchant vessels. International Archives of Occupational and Environmental Health, 61, 51-57.

- Folkard, S., Barton, J., Costa, G., Smith, L., Spelten, E., & Totterdell, P. (1993). The Standard Shiftwork Index. Ergonomics, 36(1-3), 313-318.
- Folkard, S., Monk, T. H., & Lobban, M. C. (1979). Towards a Predictive Test of Adjustment to Shift Work. Ergonomics, 22(1), 79-91.
- Härmä, M., Ilmarinen, J. & Knauth, P., (1988). Physical Fitness and Other Individual Factors Relating to Shiftwork Tolerance of Women. Chronobiology International, 5(4), 417-424.
- Howarth, H. D., Pratt, J. H., & Tepas, D. I. (1999). Do Maritime Crew Members Have Sleep Disturbances? International Journal of Occupational and Environmental Health, 5, 95-100.
- Johnson, L. C., & Spinweber, C. L. (1982). Quality of Sleep and Performance in the Navy: A Longitudinal Study of Good and Poor Sleepers. Naval Medical Research Center, Report No. 82-8; San Diego, CA.
- Kaliterna, L., Vidacek, S., Prizmic, A., Radošević-Vidacek, B. (1995). Is tolerance to shiftwork predictable from individual difference measures? Work & Stress, 9(2/3), 140-147.
- Kleitman, N. (1949). The sleep-wakefulness cycle of submarine personnel. In Human factors in undersea warfare (pp 329-341). Washington, DC: National Research Council.

- Knauth, P., Plett, R., Condon, R., DeVol, D., Fletcher, N., Eickoff, S., & Ottman, W. (1986). Sleep Duration and Subjective Ratings of Sleep Quality Determined by a Permanent Watchkeeping System. Night and Shiftwork. Haider (Ed.), 293-300.
- Mahan, R.P., Carvalhais, A. B., & Queen, S. E. (1990). Sleep Reduction in Night-Shift Workers: Is it sleep deprivation or a sleep disturbance disorder? Perceptual and Motor Skills, 70, 723-730.
- Patkai, P. (1971). The diurnal rhythm of adrenaline secretion in subjects with different working habits. Acta Physiologica Scandinavica, 81, 30-34.
- Parkes, K. R. (1994). Sleep Patterns, shiftwork, and individual differences: a comparison of onshore and offshore control-room operators. Ergonomics, 37(5), 827-844.
- Rutenfranz, J. (1982). Occupational Health Measures for Night- and Shift-Workers. Journal of Human Ergology, 11, 67-86.
- Sobel, M. E. (1982). Asymptotic confidence intervals for indirect effects in structural equation models. In S. Leinhardt (Ed.), Sociological Methodology 1982 (pp. 290-312). Washington DC: American Sociological Association.
- Tepas, D. I. (1982). Adaptation to Shiftwork: Fact or Fallacy? Journal of Human Ergology, 11, Suppl:1-12.

Thiis-Evensen, E. (1969). Shift work and health. In Å. Swenesson (Ed.) Studia Laboris et Salutis, Vol. 4., 81-83.

Verhaegen, P., Maasen, A., & Meers, A. (1981). Health Problems in Shiftworkers. L. C. Johnson, D. I. Tepas, W. P. Colquhoun, & M. J. Colligan (Eds.), Biological Rhythms, Sleep and Shiftwork, Advances in Sleep Research (Vol. 7pp. 271-282). New York: Spectrum.